

Murtuza Shaikh

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EDUCATION

Indian Institute of Information Technology, Gwalior

B.Tech, Mathematics and Scientific Computing | CGPA: 7.79

Aug 2023 – Jul 2027

Gwalior, India

SKILLS

- **Languages:** Python, C++, C, SQL
- **ML / Research:** LLMs & VLMs, Retrieval-Augmented Generation, AI Agents, Reinforcement Learning, NLP, Computer Vision, Deep Learning, Probability & Statistics, Experiment Design and Evaluation
- **Tools & Infrastructure:** PyTorch, TensorFlow, Hugging Face, LangChain, scikit-learn, NumPy, Pandas, OpenCV, AWS Bedrock, Docker, PostgreSQL

RESEARCH PUBLICATIONS

- **M. Shaikh et al.** “So, How Much Do LLMs Hallucinate on Low-Resource Languages? A Quantitative and Qualitative Analysis.” *GlobalSouthML Workshop, ICML 2026.*
- **M. Shaikh et al.** “HiBias: A Nine-Faceted Bias Annotation Dataset for Media Bias Detection in Hindi.” *GlobalSouthML Workshop, ICML 2026.*
- **M. Shaikh et al.** “FIND-IT! — Multimodal Fuzzy Fake Indian News Detection using Images and Text.” *Best Paper, Intl. Conf. of Mathematical and Computational Frontiers, 2025.*

RESEARCH PROJECTS

- **ARC White-Box Estimation Challenge 2026 (WhestBench) — Phase 2** Aug 2026 – Present
Tools: [Python, flopscope, NumPy, PyTorch, Hermite/Mehler Expansions, Gaussian Moment Propagation] 
 - Predicted expected post-ReLU activations of He-initialised 1024×16 ReLU MLPs **analytically from weights alone**, without sampling, under a hard FLOP budget (2^{41}).
 - Derived the general **Hermite coefficient for the Mehler expansion** of $\text{Cov}(\text{ReLU}(z_i), \text{ReLU}(z_j))$, showing the shipped “gain” closure is only its $k=1$ term; verified against brute-force sampling.
 - Localised the residual error via **oracle injection**: supplying the true per-layer mean improved MSE $19.7\times$ while the true covariance made it *worse* ($0.65\times$), proving all error lived in the mean path; corrected it with a greedy per-layer non-Gaussianity fit.
 - Scores **$1.31e-07$ on 50 held-out MLPs — $3.1\times$ better than the covariance-propagation baseline, $9.0\times$ better than Monte Carlo** — at 2.36% of budget. Documented negative results: an Edgeworth κ_4 correction failed even with oracle kurtosis, as ReLU’s kink breaks the expansion.
- **VLM Auto-Adjudication Pipeline —Runner Up at NHA AB PM-JAY (IISc Healthcare Hackathon)** Apr – May 2026
Tools: [Python, Gemma 3 12B VLM, PyMuPDF, Prompt Engineering, Evaluation Design] 
 - Built a per-page **vision-language classification and clinical-evidence extraction pipeline** for insurance claim documents across four PM-JAY package codes, scored live against a held-out organizer evaluator.
 - Designed a **two-pass VLM architecture** (document-type + entity pass, then per-package clinical-field pass) with disk-backed caching, cutting re-runs to zero fresh tokens and holding total usage at **11% of the 25.5M token budget**.
 - Ran controlled ablations over fusion strategies and found **intersection of keyword and VLM evidence** beat either signal alone (clinical F1 0.757 $\rightarrow$ **0.791**); ablated four document-ranking schemes to lift rank score from 0.008 to **0.107**.
 - Final submission: **mandatory F1 0.915**, clinical F1 0.791, extra F1 0.768, **aggregate 0.766 — Runner-Up out of 40 teams**.

EXPERIENCE

Fibe (EarlySalary Services Pvt. Ltd.)

Data Science Intern — Agentic Income & Obligation Identification

May 2026 – Jul 2026

Pune, India (On-Site)

- Built an end-to-end **agentic LLM pipeline on AWS Bedrock** classifying raw bank-statement transactions into **four income streams and two obligation types**, replacing a keyword rule engine that detected salary only.
- Designed per-category signal stacks with hard rejection filters — employer-name fuzzy matching, monthly-recurrence and amount-stability tests, and a 60+ lender library — eliminating high-cost false positives such as loan disbursements scored as income; added a **self-learning feedback loop** promoting repeated keywords into persistent libraries.
- Evaluated on **15,758 customers**: detected salary for **40.6%**, incl. **7.0% missed by both the rule engine and human underwriters**; surfaced business (20.7%), government-scheme (6.8%), EMI (12%) and rent (3%) signals with zero prior coverage, raising the underwritable base to **52.1%** and correcting systematically optimistic FOIR.
- Productionized at **~ 0.8 s and 1.7 per customer** (80 customers/min, 40 concurrent workers), cheap enough to run on the full applicant base rather than a sample.

ACHIEVEMENTS

- **Winner, Smart India Hackathon 2024** — 1st of 500 teams nationwide (multimodal conversational chatbot for Bharat Electronics Ltd.) Dec 2024
- **Runner-Up, AB PM-JAY Auto-Adjudication Hackathon**, NHA & IISc Bangalore — 2nd of 40 teams May 2026
- **Special Mention, AMD AI Hackathon** (AMD, Unsloth, IIT Delhi) — 4th of 60 teams; GRPO-trained RL agent for Minesweeper Feb 2026
- **Finalist, Meta RL Hackathon** — top 800 of 30,000 teams 2026
- **CASML-2024 Scientific ML Hackathon**, IISc Bangalore — 15th of 47 teams; PINN for urban chemical dispersion Dec 2024
- **JEE Advanced 2023:** AIR 9448 | **JEE Mains 2023:** AIR 10314 (99.12 percentile) 2023